Table 1: Effect of Ownership Change on Nursing Staffing

Outcome	RN	LPN	CNA	Total
HPRD	-0.0368^{***}	-0.0039	-0.0532^{***}	-0.0940^{***}
	(0.0055)	(0.0062)	(0.0096)	(0.0134)
Log(HPRD)	-0.1174^{***}	-0.0022	-0.0270^{***}	-0.0293^{***}
	(0.0168)	(0.0095)	(0.0049)	(0.0041)
Hours	-80.0180^{***}	68.4756^{***}	64.8807^{*}	53.3383
	(15.3077)	(17.8010)	(34.9039)	(49.9070)
Log(hours)	-0.0767^{***}	0.0384^{***}	0.0133^{*}	0.0110^{*}
	(0.0173)	(0.0103)	(0.0068)	(0.0060)

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

HPRD is hours per resident day; hours are the HPRD numerator, reported to test whether the HPRD decline is mechanically driven by the occupancy-rate increase in the denominator.

The anticipation window ($\tau = -3, -2, -1$) is excluded.

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 2: Effect of Ownership Change on Business-Model Outcomes

Outcome	Coefficient (SE)	Observations
Occupancy rate	2.6841***	550,330
	(0.3497)	
Spare capacity	-0.0263^{***}	550,330
	(0.0035)	
Medicare share	1.6100***	550,330
	(0.2159)	
Medicaid share	-1.1710^{*}	550,330
	(0.5952)	
Average length of stay	-13.3641^{***}	540,346
	(4.3491)	
Case mix	0.0525***	457,842
	(0.0059)	

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

Occupancy rate is residents as a share of available bed-days; spare capacity is unused certified beds as a share of certified beds. Payer shares are shares of patient days. The anticipation window ($\tau = -3, -2, -1$) is excluded.

Case mix is restricted to 2017/10–2023/12. CMS changed its case-mix methodology in October 2017 and January 2024; both breaks shift the level and the cross-sectional spread simultaneously across the whole sample, which facility and period fixed effects do not absorb.

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 3: Effect of Ownership Change on Quality Measures

Outcome	(1)	(2)	Observations
Panel A: Long-stay labor-saving mechanism measures			
Catheter use	-0.1695^{***}	-0.1711^{***}	172,671
	(0.0488)	(0.0488)	
Anti-psychotic medication use	-0.3362	-0.3543	171,358
	(0.2218)	(0.2231)	
Anti-anxiety/hypnotic medication use	-0.2321	-0.2118	171,854
	(0.2019)	(0.2027)	
Panel B: Long-stay resident outcome measures			
Pressure injuries	0.2586	0.2825	121,962
	(0.1762)	(0.1783)	
Falls with major injury	0.0125	0.0138	175,123
	(0.0636)	(0.0644)	
Weight loss	0.2042^{*}	0.2080^{*}	170,733
	(0.1040)	(0.1034)	
Decline in physical functioning	0.3240	0.3501	163,983
	(0.2695)	(0.2782)	
Urinary tract infections	-0.3279^{***}	-0.3181^{***}	174,381
	(0.0765)	(0.0767)	
Panel C: Short-stay measures			
New antipsychotic medication	-0.0187	-0.0219	134,020
	(0.0432)	(0.0433)	
Improved function	-0.0569	-0.0880	88,792
	(0.4888)	(0.4889)	

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

Column (1) is the baseline specification. Column (2) adds RN, LPN, and CNA hours per resident day as controls. Because staffing is itself affected by ownership change, column (2) is not a preferred estimate of the total effect; it is reported to show whether the quality response is attenuated after conditioning on measured staffing inputs.

Long-stay measures (Panels A-B) and short-stay measures (Panel C) are constructed from different resident populations and are not directly comparable to one another. For every measure, lower values indicate better measured quality.

Pressure injuries is estimated on 2018–2023 only; improved function is estimated on 2017–2022 only. Both measures show near-complete absence outside these windows and are trimmed to the years where they are actually reported rather than treated as full-panel outcomes. Vaccination measures are excluded from this table.

The transition quarter ($\tau = 0$) is excluded. Observations are reported for column (1); column (2) drops facility-quarters with missing staffing.

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 4: Effect of Ownership Change on Non-Nurse (Therapy) Staffing

Staff type	Coefficient (SE)	Observations
Physical Therapist (PT)	-0.0002	550,330
	(0.0017)	
PT Assistant	0.0012	550,330
	(0.0020)	
PT Aide	0.0047^{***}	550,330
	(0.0008)	
Occupational Therapist (OT)	-0.0024	550,330
	(0.0016)	
OT Assistant	0.0039^{**}	550,330
	(0.0018)	
OT Aide	0.0029^{***}	550,330
	(0.0005)	
Speech-Language Pathologist	-0.0031^{***}	550,330
	(0.0010)	
Total Non-Nurse	0.0070	550,330
	(0.0059)	

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

Outcomes are hours per resident day (HPRD) for each non-nurse staff type. Levels only are reported: several categories (PT Aide, OT Aide) have a large share of exact zeros, which makes log specifications and a raw-hours decomposition less informative here than for nursing staff.

The anticipation window ($\tau = -3, -2, -1$) is excluded.

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.